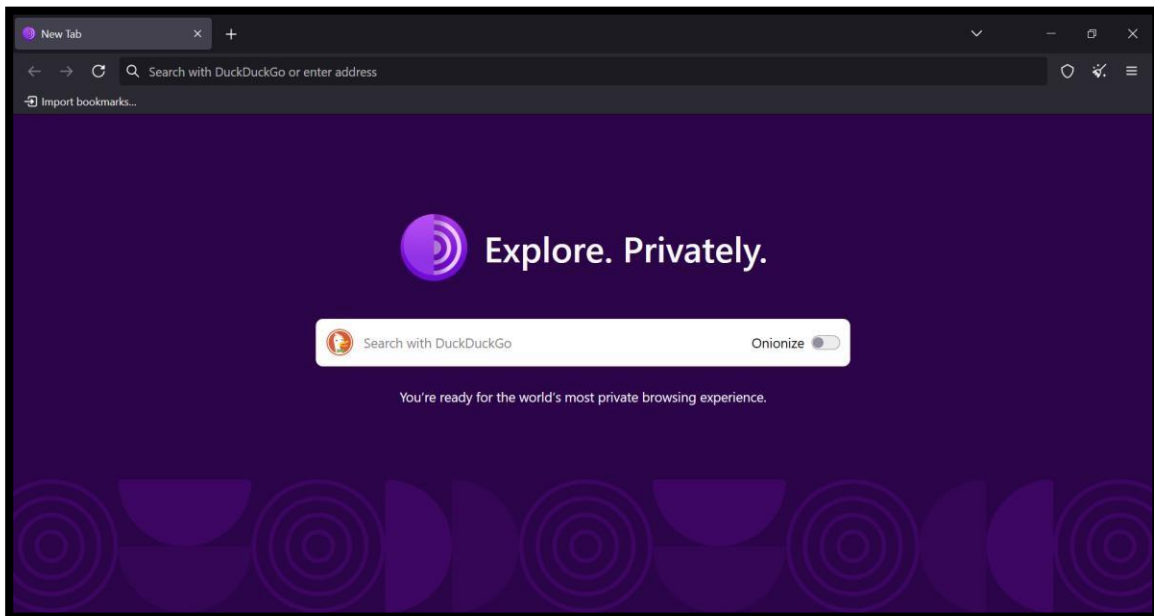


Lab 9: Tor Browser Forensics - Memory Acquisition and Artifacts

Objective

The objective of this lab is to identify and analyze traces left by the Tor browser on a Windows-based system for forensic investigation. This includes examining memory dumps, Windows registry entries, command prompt artifacts, state files, and prefetch



files.

1. Command Prompt Analysis

The Windows Command Prompt can be used to investigate running processes, network connections, and file system changes related to the Tor browser.

- Check Running Processes

```
C:\Windows\System32>tasklist | findstr /i "tor"
AggregatorHost.exe           6364 Services           0      16,968 K
tor.exe                      14596 Console            1      89,164 K
```

tasklist | findstr /i "tor": Lists all running processes and filters for "tor".

- Check Network Connections

```
C:\Windows\System32>netstat -ano | findstr ":9050"
C:\Windows\System32>
```

netstat -ano | findstr ":9050": Displays active network connections, filtering for Tor's port (9050). Tor typically uses port **9050**. If there's an active connection, it indicates that the Tor browser is running or was recently used.

- Search for Tor Browser Files

```
C:\Windows\System32>dir /s /a "C:\Users\%USERNAME%\Desktop\Tor Browser"
Volume in drive C is Local Disk
Volume Serial Number is 40AE-B759

Directory of C:\Users\91992\Desktop\Tor Browser

22-03-2025  23:17    <DIR>          .
22-03-2025  23:18    <DIR>          ..
22-03-2025  23:17    <DIR>          Browser
22-03-2025  23:17                812 Tor Browser.lnk
                        1 File(s)                812 bytes

Directory of C:\Users\91992\Desktop\Tor Browser\Browser

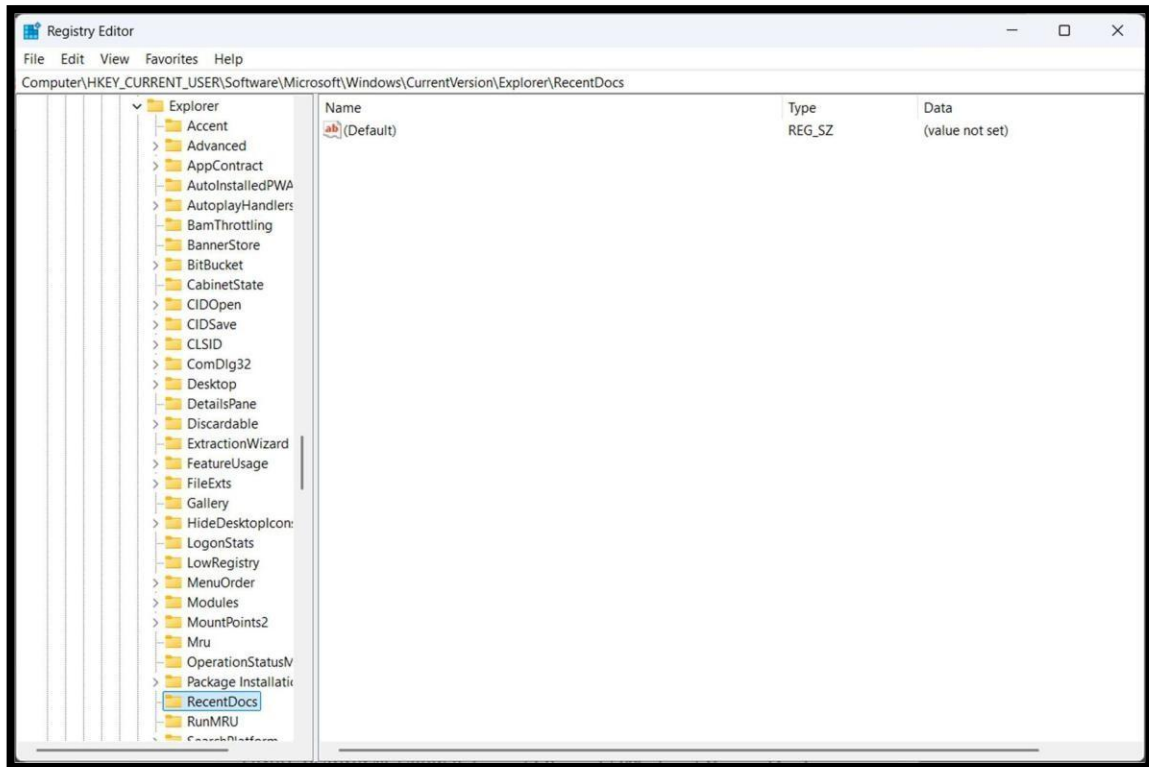
22-03-2025  23:17    <DIR>          .
22-03-2025  23:17    <DIR>          ..
03-03-2025  15:07                26,112 AccessibleMarshal.dll
03-03-2025  15:07                660 application.ini
22-03-2025  23:17    <DIR>          browser
03-03-2025  15:07       4,410,176 d3dcompiler_47.dll
22-03-2025  23:17    <DIR>          defaults
03-03-2025  15:07                55 dependentlibs.list
22-03-2025  23:17    <DIR>          distribution
03-03-2025  15:07       1,758,208 firefox.exe
03-03-2025  15:07                557 firefox.VisualElementsManifest.xml
22-03-2025  23:17    <DIR>          fonts
03-03-2025  15:07       1,003,008 freebl3.dll
03-03-2025  15:07      10,488,832 gkcodecs.dll
03-03-2025  15:07       328,192 ipccclientcerts.dll
03-03-2025  15:07       504,832 lgpllibs.dll
03-03-2025  15:07       814,592 libEGL.dll
03-03-2025  15:07       6,119,936 libGLSv2.dll
03-03-2025  15:07       4,433,920 mozavcodec.dll
03-03-2025  15:07       654,336 mozavutil.dll
03-03-2025  15:07      1,430,528 mozglue.dll
03-03-2025  15:07       784,384 nmhproxy.exe
03-03-2025  15:07       2,716,672 nss3.dll
03-03-2025  15:07       491,520 nssckbi.dll
03-03-2025  15:07      19,490,631 omni.ja
```

dir /s /a "C:\Users\%USERNAME%\Desktop\Tor Browser": Searches for the Tor Browser folder in the user's directory.

2. Registry Analysis

Windows Registry stores traces of installed and executed applications.

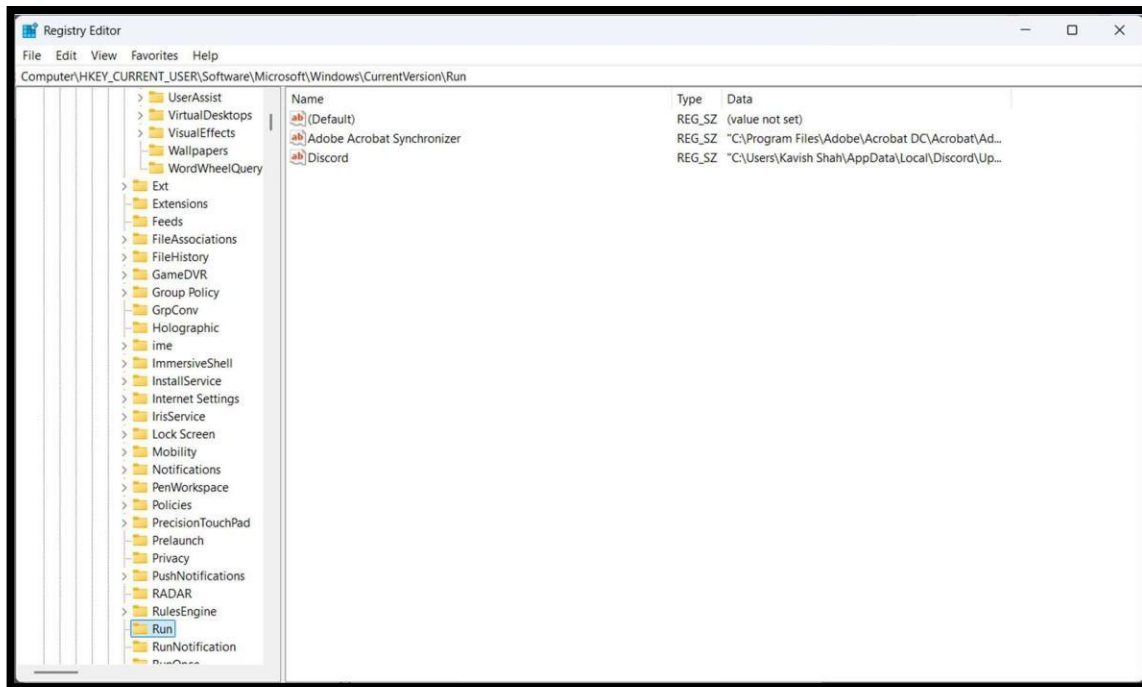
- Checking Recently Used Programs:
 - Open Registry Editor (regedit.exe) and navigate to:
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
 - Look for tor.exe or any related files.



- Analyzing Run Keys:

Navigate to:

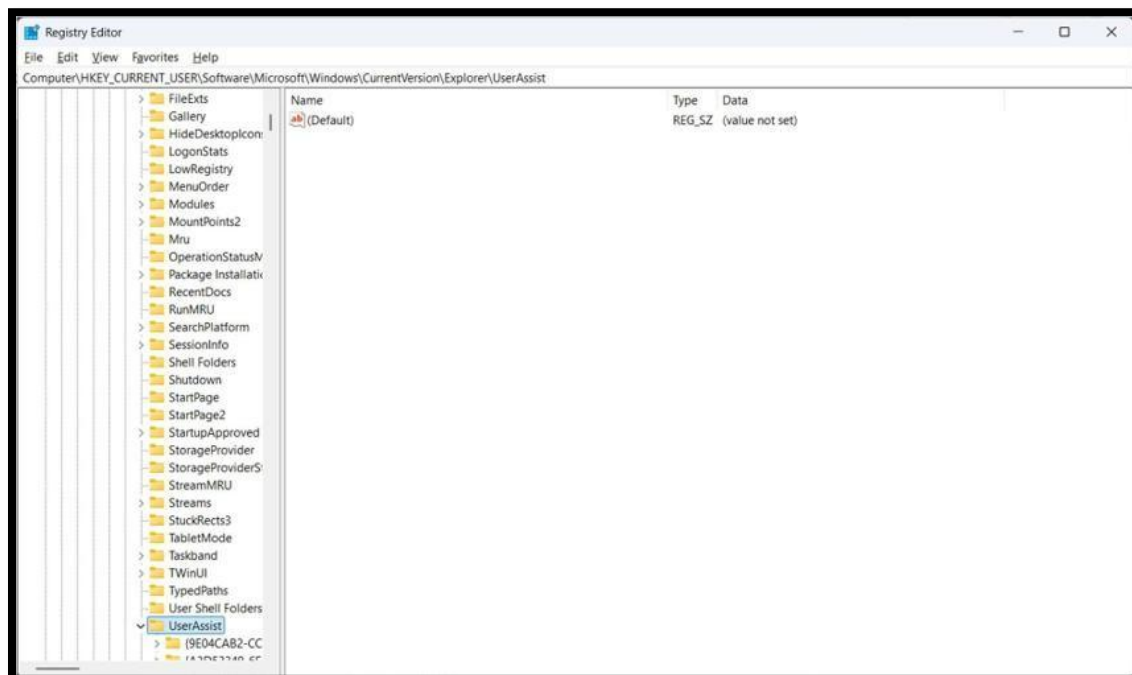
 - HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
 - If tor.exe is listed, it indicates persistence.

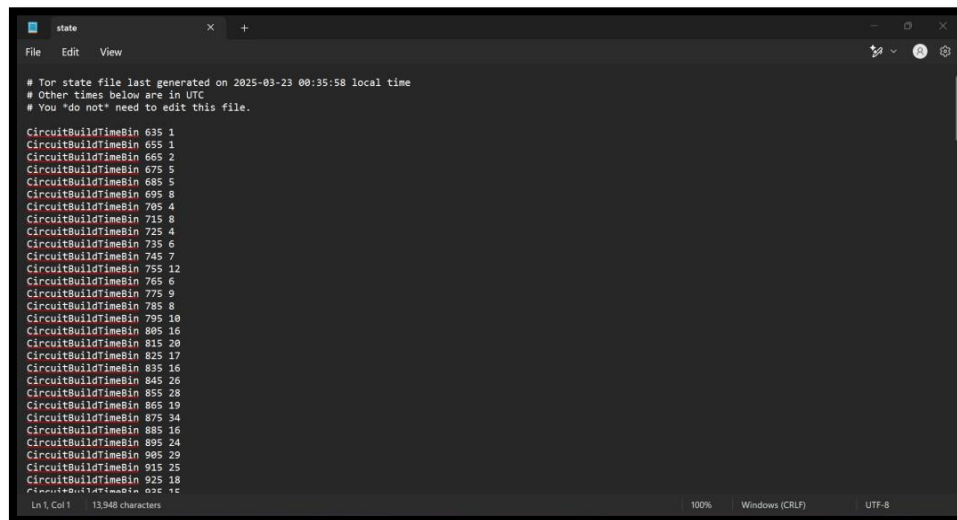


- Examining UserAssist Entries:

Use the following path:

- HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist
- Look for encoded values related to Tor activity.





```
# Tor state file last generated on 2025-03-23 00:35:58 local time
# Other times below are in UTC
# You "do not" need to edit this file.

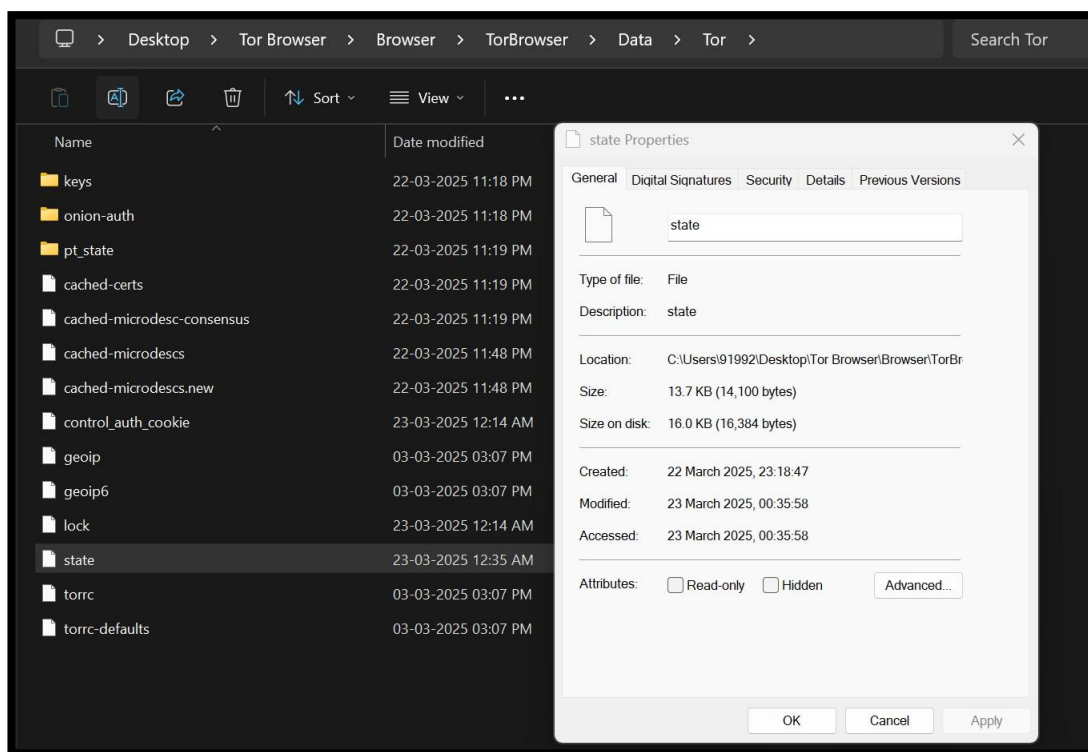
CircuitBuildTimeBin 635 1
CircuitBuildTimeBin 655 1
CircuitBuildTimeBin 665 2
CircuitBuildTimeBin 675 5
CircuitBuildTimeBin 685 5
CircuitBuildTimeBin 695 8
CircuitBuildTimeBin 705 4
CircuitBuildTimeBin 715 8
CircuitBuildTimeBin 725 4
CircuitBuildTimeBin 735 6
CircuitBuildTimeBin 745 7
CircuitBuildTimeBin 755 12
CircuitBuildTimeBin 765 6
CircuitBuildTimeBin 775 9
CircuitBuildTimeBin 785 8
CircuitBuildTimeBin 795 10
CircuitBuildTimeBin 805 15
CircuitBuildTimeBin 815 20
CircuitBuildTimeBin 825 17
CircuitBuildTimeBin 835 16
CircuitBuildTimeBin 845 26
CircuitBuildTimeBin 855 28
CircuitBuildTimeBin 865 19
CircuitBuildTimeBin 875 34
CircuitBuildTimeBin 885 16
CircuitBuildTimeBin 895 24
CircuitBuildTimeBin 905 29
CircuitBuildTimeBin 915 25
CircuitBuildTimeBin 925 18
CircuitBuildTimeBin 935 16
```

State File Analysis

- Locate Tor State Files:

Navigate to the Tor profile directory:

- C:\Users\<username>\AppData\Roaming\Tor\
- Look for state files that contain Tor's last known connection status.



- Analyzing Content:

Open the state file in a text editor and examine details such as:

- Last accessed relays.
- Bridges used.
- Circuit IDs.

- Extracting Timestamps: Identify timestamps indicating when Tor was last used.

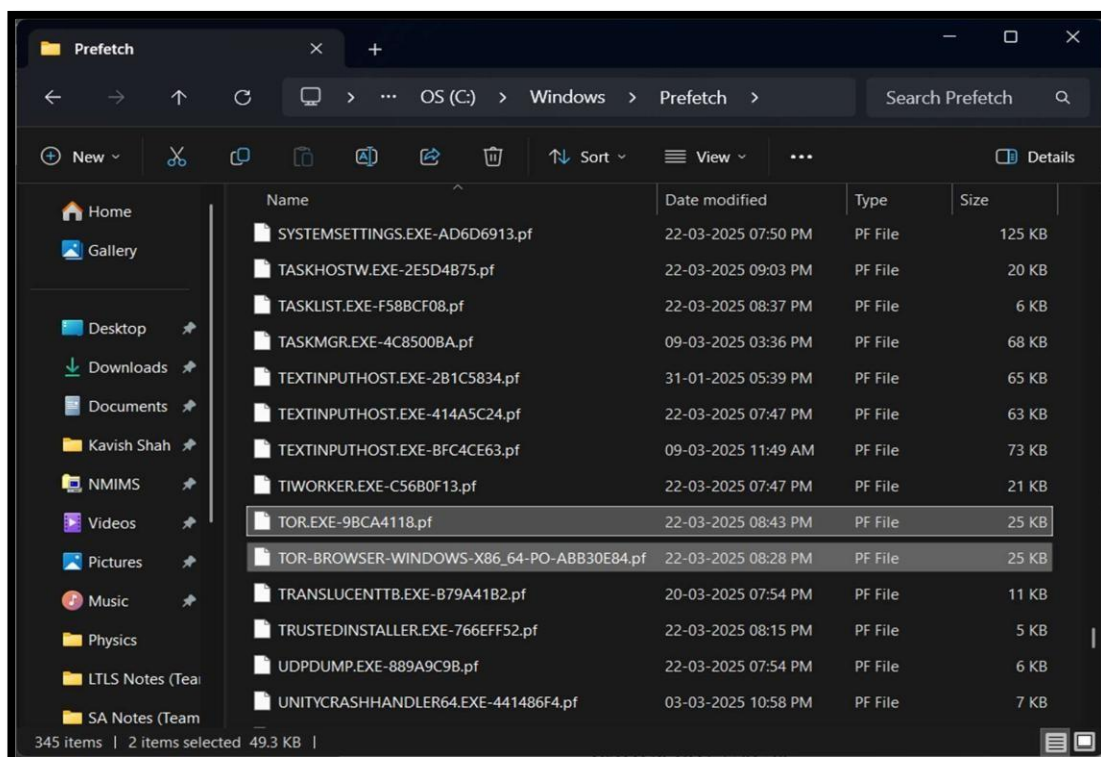
```

state
File Edit View
CircuitBuildTimeBin 6235 1
CircuitBuildTimeBin 7105 1
CircuitBuildAbandonedCount 504
Dormant 0
Guard in=default rsa_id=48A87F81823A1928A61F90543205E620F2846CE0 nickname=myFirstMiddleRelay sampled_on=2025-03-22T17:44:43 sampled_idx=0 sampled_by=0.4.8.14 listed=1
confirmed_on=2025-03-19T18:07:58 confirmed_idx=0 pb_circ_attempts=26.000000 pb_circ_successes=26.000000 pb_successful_circuits_closed=17.000000 pb_collapsed_circuits=
9.000000
Guard in=default rsa_id=ACB898B28ACABBD78F23AF2F806FC864A411646 nickname=FTPhalifaxRWH sampled_on=2025-03-17T15:53:48 sampled_idx=1 sampled_by=0.4.8.14 listed=1
confirmed_on=2025-03-17T08:47:47 confirmed_idx=1 pb_use_attempts=3.000000 pb_use_successes=3.000000 pb_circ_attempts=29.000000 pb_circ_successes=29.000000
pb_successful_circuits_closed=26.000000 pb_collapsed_circuits=3.000000
Guard in=default rsa_id=BFC404411908008E8E70218978EE83C3F16AE7A nickname=antifaang sampled_on=2025-03-11T12:21:08 sampled_idx=2 sampled_by=0.4.8.14 listed=1
confirmed_on=2025-03-17T15:10:03 confirmed_idx=2 pb_circ_attempts=21.000000 pb_circ_successes=20.000000 pb_successful_circuits_closed=20.000000
Guard in=default rsa_id=D4380F8801539CE1A94F902F4141F5916DAF8B3 nickname=mittig2 sampled_on=2025-03-16T05:23:06 sampled_idx=3 sampled_by=0.4.8.14 listed=1 confirmed_on=
2025-03-15T18:44:55 confirmed_idx=3
Guard in=default rsa_id=409FD80A3CEAA4FF20F488FA7F52AB00E4338F56 nickname=arbitraryTessa4 sampled_on=2025-03-22T13:05:42 sampled_idx=4 sampled_by=0.4.8.14 listed=1
confirmed_on=2025-03-16T20:18:59 confirmed_idx=4
Guard in=default rsa_id=B8089EC4MC93928711A0806080F547A09AEF2D67 nickname=bauruine sampled_on=2025-03-19T12:17:13 sampled_idx=5 sampled_by=0.4.8.14 listed=1 confirmed_on=
2025-03-14T19:27:58 confirmed_idx=5
Guard in=default rsa_id=3380B8798E2FFA55797DF05218CF5CD57AA7F1AF nickname=bauruine sampled_on=2025-03-20T22:20:18 sampled_idx=6 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=D021D70CD592D18DEFB3B10F787746983B350FFE nickname=TorRelay3c sampled_on=2025-03-20T05:01:03 sampled_idx=7 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=0AE076CD2A42018F2D98C325211A10D2C8006CE9 nickname=Tajjina sampled_on=2025-03-15T07:55:41 sampled_idx=8 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=6ABD2C9C94A3D3A8BC413341DA3D1DC10A52C nickname=trm2 sampled_on=2025-03-21T17:40:40 sampled_idx=9 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=ACGAF012A4A940914DC441746B9342A6EC116417 nickname=Unnann sampled_on=2025-03-21T03:35:29 sampled_idx=10 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=FC8B8AF5A54FF75F81FA928109887345EE83D03 nickname=1blu4DEicebeer01 sampled_on=2025-03-16T08:22:39 sampled_idx=11 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=E870EDFEC5A2B1C3867434F9FB726FA4154833A nickname=lagakity2 sampled_on=2025-03-14T07:36:31 sampled_idx=12 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=7302F8AAC76835768AF68849CA07D1CC38C5E nickname=OceanGhost sampled_on=2025-03-21T23:45:19 sampled_idx=13 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=D95A2DF178158A52E4EA4902D46EBAEB1DCEBABC nickname=0x90 sampled_on=2025-03-12T03:50:41 sampled_idx=14 sampled_by=0.4.8.14 listed=1
Guard in=default rsa_id=1024292113610F0096FC5546E4EC338E27A4F54 nickname=bauruine sampled_on=2025-03-22T11:25:31 sampled_idx=15 sampled_by=0.4.8.14 listed=1
confirmed_on=2025-03-13T20:08:02 confirmed_idx=7 pb_circ_attempts=262.750000 pb_circ_successes=260.500000 pb_successful_circuits_closed=18.000000 pb_collapsed_circuits=
242.500000 pb_timeouts=244.000000
Guard in=default rsa_id=973E3402658CD70098311642776DFBC576ACE1E nickname=Abaddon sampled_on=2025-03-12T00:23:57 sampled_idx=17 sampled_by=0.4.8.14 listed=1 confirmed_on=
2025-03-14T19:48:37 confirmed_idx=6 pb_circ_attempts=4.000000 pb_circ_successes=4.000000 pb_successful_circuits_closed=4.000000
Guard in=default rsa_id=E083B10226F067A1B215178019A19F31E7FDC3 nickname=KherN1 sampled_on=2025-03-16T15:39:41 sampled_idx=18 sampled_by=0.4.8.14 listed=1 confirmed_on=
Ln 1, Col 1 13,866 characters 90% Windows (CRLF) UTF-8

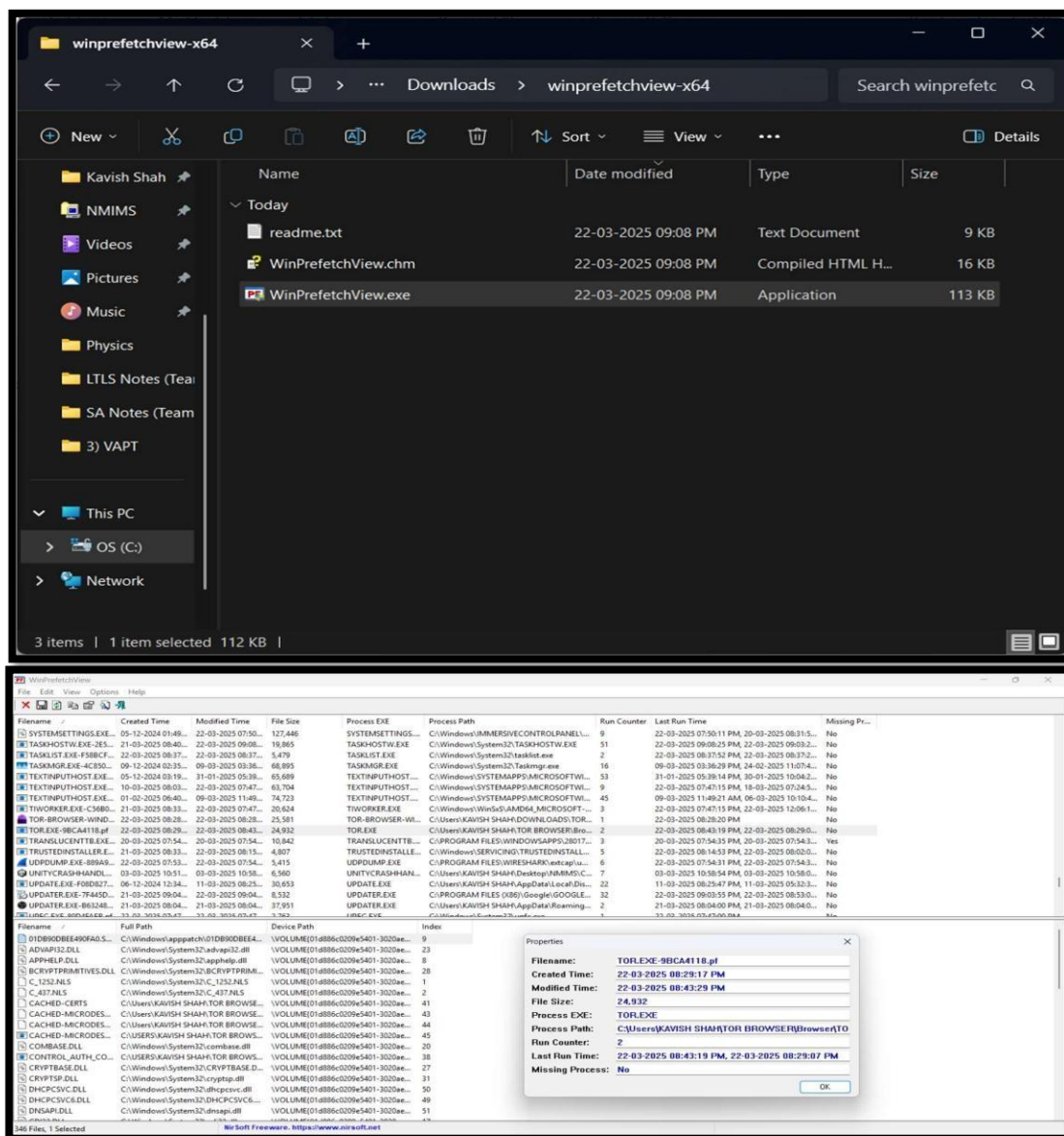
```

3. Prefetch Files

- Locating Prefetch Entries:
Navigate to:
 - C:\Windows\Prefetch\
 - Look for TOR.EXE-XXXXXXX.pf.



- Extracting Execution Details:
Prefetch files contain:
 - First and last execution timestamps.
 - Number of times Tor was executed.
- Using Third-Party Tools: Use WinPrefetchView or PECmd.exe to parse prefetch files.



Conclusion: This lab demonstrates key forensic techniques to identify traces of the Tor browser on a Windows system. By analyzing command prompt outputs, registry entries, state files, and prefetch data, investigators can gather evidence of Tor usage, including execution timestamps, network connections, and persistence attempts. These artifacts provide valuable insights into user activity, even after Tor is closed. Proper forensic tools and methodologies are crucial for preserving evidence and understanding potential anonymity-related threats.